

Functional Analysis — Homework 3

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Problem 1. Let X, Y be normed spaces and $\mathcal{B}(X, Y)$ the vector space of all bounded linear operators $T : X \rightarrow Y$. Show that the operator norm

$$\|T\|_{op} := \sup_{\|x\| \leq 1} \|T(x)\|$$

is a norm in $\mathcal{B}(X, Y)$. Show that if Y is Banach, then $\mathcal{B}(X, Y)$ is Banach.

Problem 2. Let X, Y be normed spaces with X infinite-dimensional and $Y \neq \{0\}$. Using the axiom of choice (or equivalent statement), show that there exists a linear operator $T : X \rightarrow Y$ which is unbounded.

Problem 3. Let $C^\infty[0, 1]$ be the vector space of all infinitely differentiable functions from $[0, 1]$ to $\mathbb{R}$. Consider the derivative as a linear operator

$$\mathcal{D} : C^\infty[0, 1] \rightarrow C^\infty[0, 1].$$

Show that it is impossible to equip $C^\infty[0, 1]$ with a norm $\|\cdot\|$ such that $\mathcal{D}$ is bounded.

Problem 4. Recall that c_{00} is dense in ℓ^1 . Consider both spaces equipped with the ℓ^1 norm. Show that the identity map

$$\begin{aligned} \iota : c_{00} &\rightarrow c_{00} \\ x &\mapsto x \end{aligned}$$

cannot be extended to a map $\tilde{\iota} : \ell^1 \rightarrow c_{00}$.

Problem 5. Let X be a Banach space and $T : X \rightarrow \ell^\infty$ a linear operator. Writing

$$T(x) = (f_1(x), f_2(x), f_3(x), \dots) \in \ell^\infty,$$

we get a sequence of linear functionals $f_n : X \rightarrow \mathbb{R}$. Show that T is bounded if and only if each f_n is a bounded linear functional on X .